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DEPARTMENT OF DEFENSE HANDBOOK

ELECTROMAGNETIC ENVIRONMENTAL EFFECTS TO ORDNANCE GUIDE

PART 4

EXTERNAL RADIO FREQUENCY ELECTROMAGNETIC ENVIRONMENTS

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FOREWORD

1. This handbook is approved for use by all Departments or Agencies of the Department of Defense (DoD).

2. This part supplements MIL-HDBK-240-1 and was prepared for contractors, DoD test agencies, activities, and test planners. This handbook is for guidance only and cannot be cited as a requirement.

3. This handbook provides recommended practices for conducting Electromagnetic Environmental Effects (E3) to Ordnance evaluations. Developed over many years by DoD test activities, the methods and procedures described herein are the product of the collective DoD radio frequency (RF) electromagnetic environment (EME) test experience. This handbook consolidates these practices into a single document vice numerous individual Service standards, instructions, and so forth, thus promoting a standardized approach.

4. This handbook was prepared by the Services under the sponsorship of the Joint Ordnance Commander's Group (JOCG) E3 Ordnance Safety Subgroup in accordance with the guidelines of the Standardization Reform Policy established by the Secretary of Defense.

5. Comments, recommendations, additions, or deletions and any other pertinent data that may improve this document should be emailed to marcus.shellman.civ@mail.mil or addressed to:

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1. SCOPE

1.1 Purpose. This part of MIL-HDBK-240 provides recommended methods and techniques to develop guidance for the assessment of weapon systems, ordnance, and associated support equipment during exposure to the external radio frequency (RF) electromagnetic environment (EME) with regard to its operational electronics. It is concerned with external RF EME evaluation for all Air Force, Army, Navy, Marine Corps, and Coast Guard ordnance systems and support equipment for all mission areas. Although intended primarily for use by Department of Defense (DoD) external RF EME test and evaluation activities, the handbook also provides a consolidation of "corporate knowledge" about the subject that should be of interest to procurement authorities and system developers.

1.2 Background. There are four specific objectives of this handbook:

a. Document external RF EME joint-service ordnance test methodology.

b. Promote test standardization.

c. Identify supplemental techniques.

d. Facilitate the exchange of external RF EME test data.

Ultimately, external RF EME test data is used to verify ordnance compliance to the external RF EME requirement, assess risks, and develop effective control measures to minimize those risks.

External RF EME assessments are essentially a determination of how ordnance items respond to the EME that could be encountered by the ordnance during their operational use. The response can be influenced by three principal factors: (1) EME characteristics, (2) physical configuration(s) of the ordnance, and (3) operational electronics. Each factor should be considered from the time the ordnance item is fielded until it completes its life cycle.

When defining the operational EME, one should anticipate all sources of the EME (the individual Service EME, the EME in joint-operation scenarios, the threat environment, and so forth). The Joint Capabilities Integration and Development System (JCIDS) Manual states that, "Because all weapons/weapon systems have the potential of being deployed together or employed in joint environments, weapons and weapon systems will be considered joint systems within the JCIDS process and may be designated as Joint Requirements Oversight Council (JROC) or the Joint Capabilities Board (JCB) Interest." As joint operations become more commonplace, there is an increased likelihood that weapons from one Service will be exposed to other Services' EME levels. MIL-STD-464 specifies the EME levels for external RF EME verification based on environments documented in MIL-HDBK-235. For all systems, the appropriate environment defined in MIL-HDBK-235 may be extracted and used for tailoring.

2. APPLICABLE DOCUMENTS

2.1 General. See MIL-HDBK-240-1. This document can be found on ASSIST Online at https://assist.dla.mil.

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3. DEFINITIONS

3.1 Definitions. See MIL-HDBK-240-1.

4. EXTERNAL RF EME EVALUATION

4.1 Overview. Ordnance and its associated support systems are employed in proximity to platforms that operate radar, communications transmitters, and electronic warfare systems. These high-powered emitters generate an external RF EME, which can result in electromagnetic interference (EMI) observed in collocated systems and equipment. EMI can be a result of both unintentional coupling of RF energy into electrical/electronic circuits (back-door coupling) and into intentional receivers (front-door coupling). The external RF EME can penetrate electrical or electronic systems at various points of entry (POE) and couple to critical components. Component responses can cause circuits to malfunction resulting in performance degradation and mission failure.

Electronic/electrical systems should survive and operate safely and reliably in the intended external RF EME. These systems should be evaluated prior to fielding to characterize system performance during exposure to the external RF EME. Analysis can be used to predict the performance of ordnance systems. However, due to the multitude of variables that affect system performance in a complex operational environment, analysis alone can be insufficient. The reality is that ordnance systems must operate in the external RF EME without failure or severe degradation that would adversely affect mission completion or compromise safety. Acquiring hard data through preliminary analysis and testing can be used to assess risk associated with the weapon systems, ordnance, and associated support equipment ability to perform its mission and meet safety requirements in its intended EME.

4.2 Approaches to Addressing External RF EME Evaluation. Due to the multitude of experiences within the Services relative to external RF EME evaluation, Service-unique approaches have evolved over time to address the issue. Aside from Service history, other factors have influenced the respective external RF EME programs, such as the determination of acceptable risk levels and mitigation methods. For example, when electrical/electronic equipment is deemed susceptible to the external RF EME, the Army and Air Force might determine the risk to be acceptable due to the low probability of the equipment being exposed to the particular EME of concern, whereas, limited space aboard surface ships might leave no other option for the Navy than to implement more rigorous modifications to mitigate susceptibilities before fielding.

4.2.1 Mission. Despite differences in the way each Service manages external RF EME evaluation, one common element is the definition of the expected external RF EME levels for system operation. A description of the mission of the ordnance and platform is needed to understand how the item will be used throughout its service life and, thus, becomes the basis for determining appropriate external RF EME test procedures. This is the basis for selecting which stockpile-to-safe separation sequence (S4) configurations and ancillary equipment should be tested. Furthermore, thorough knowledge of the mission is helpful in selecting the test EME and determining appropriate procedures for exercising the ordnance system under test (SUT). MIL-STD-464 provides worst-case EME levels for systems used in different applications, including:

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shipboard weather deck/flight deck, shipboard main-beam, space and launch vehicle, ground, rotary wing aircraft, and fixed-wing aircraft operations. The external RF EME levels defined in MIL-STD-464 can be tailored to the system's expected operational EME. Additional details regarding specific operational EMEs can be found in MIL-HDBK-235.

4.2.2 Program Responsibilities. There are a number of activities within each Service that are related to external RF EME evaluation. Activities with assigned technical roles generally are concerned with determining whether the system design is adequate to satisfy the electromagnetic environmental effects (E3) requirements of MIL-STD-464. This most often is accomplished via Government-sponsored tests at Government-operated test facilities. Table I identifies key Army, Navy/Marine Corps, and Air Force test agencies that perform external RF EME testing.

TABLE I. DoD external RF EME test agents

Service	Facility	Location
Army	Redstone Test Center	Redstone Arsenal, AL
	White Sands Test Center	White Sands Missile Range, NM
	Aberdeen Test Center	Aberdeen Proving Ground, MD
	Electronic Proving Ground	Ft. Huachuca, AZ
Navy & Marine Corps	Naval Surface Warfare Center, Dahlgren Division	Dahlgren, VA
	Naval Air Warfare Center, Aircraft Division	Patuxent River, MD
	Naval Air Warfare Center, Weapons Division	China Lake, CA
Air Force	Joint Preflight Integration of Munitions and Electronic Systems	Eglin AFB, FL
	Edwards AFB*	Edwards AFB, CA
	Air Force Life Cycle Management Center	Wright-Patterson AFB, OH

* AFB = Air Force Base.

4.3 External RF EME Assessments. The purpose of the external RF EME evaluation is to demonstrate that ordnance systems will not be adversely affected when exposed to their operational EME. Since system responses to the external RF EME are highly dependent on the system test configuration, significant effort should be put forth in configuring test assets and test setups in order to obtain data that is an accurate reflection of the tactical system. Furthermore, when the design of the ordnance and/or platform is altered or it is desirable to leverage external RF EME data from another system, it is critical that a detailed review of the designs be conducted before determining if the test data are applicable. External RF EME assessments may also employ preliminary analysis as part of the approach for assessing ordnance system performance to predict system responses and coupling mechanisms that can be exploited during testing.

4.3.1 Analysis. Preliminary analysis may include modeling and simulation (M&S) and/or review of prior test results including enclosure shielding effectiveness (SE) test data,

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equipment and subsystem level EMI test data, and prior system-level test data. The use of commercially available M&S software can be used to evaluate coupling methods, resonant frequencies, and POEs, allowing the system developer to identify system sensitivities over a wide band of EME frequencies prior to hardware implementation. Analysis of SE test data for ordnance/platform enclosures can be used to validate design goals and establish EME test levels for subsystems and equipment housed within the enclosure. Additional analysis of equipment and subsystem-level EMI testing for radiated and conductive susceptibility as detailed in MIL-STD-461 can identify system circuit sensitivities to specific EME parameters such as frequency, power density, polarization, and modulation. Lastly, prior results from system-level external RF EME testing establish the baseline system performance for future changes in ordnance and/or platform configuration.

Analytical methods are limited in their ability to provide complete system-level evaluation in the presence of the external RF EME. For example, present M&S methods are unable to resolve complex coupling paths and are thus limited in their ability to provide complete system-level evaluation in the presence of the external RF EME. Hence, M&S should not be used solely for qualification. Analytical methods generally are used as preliminary assessments to identify those ordnance configurations, procedures, and EME parameters needing emphasis during testing. In some cases, analyses may provide sufficient justification to recommend evaluation via similarity. However, experience has shown that even minor changes can have unexpected effects and, thus, testing is generally preferred.

4.3.2 Test. A purely analytical approach for predicting system response is generally not recommended because analysis alone generally results in an overly restrictive assessment. Therefore, testing should remain an essential element of each Service's E3 program. However, testing has its own limitations. One drawback is the practical limitation on the number of independent samples that can be obtained in the timeframe of the test event. For example, there are limits on the number of frequencies that can be used to simulate the operational EME. The result of an inadequate number of frequency samples can be a failure to capture the maximum responses or resonant frequencies within a given frequency band. Despite the imperfections of the process, testing is generally recognized as the preferred means of determining expected system performance within specified EME levels. For example, analytical models can be used during test preparation to predict system resonant frequencies and orientations, which then can be investigated during the test.

5. EXTERNAL RF EME TEST PLANNING

5.1 Overview. The success of an external RF EME test event is highly dependent on the rigor applied to the test planning process. Proper test planning requires that certain key information be obtained about the SUT. The process begins with a complete identification of the ordnance item, including all components/subcomponents, host platform(s), any equipment with which it interfaces, all modes of operation, and the system's physical characteristics. This information then is analyzed to determine how the test should be conducted; for example, how should the system be configured for testing, what modes/features should be evaluated, what EME parameters should be used to represent the tactical EME, and what criteria/metrics should be used to assess system performance during the test. In essence, the planning phase consists of gathering and evaluating information. An engineering evaluation of the information should be

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made to develop a plan for accomplishing the test objectives in the most efficient and effective manner. The following paragraphs provide guidance for developing such a plan.

5.2 Defining the Test Environment. One of the first priorities is to develop a test EME that is a realistic representation of the system's intended operational EME. Through research, it should be determined where the system is likely to be used and what sources will be operating around the SUT. The tables in MIL-STD-464 typically represent the worst-case peak and average EME levels for different frequency bands relative to a general operational platform/area. Note that the specified peak and average field strengths detailed in the tables for a given band are not correlated to each other via a modulation factor (modulation is not defined); the peak and average field strength values specified for a given band could be representative of different emitters. Also, the EME tables in MIL-STD-464 should be considered as default values for determining a test EME when better data used for tailoring an EME is not available.

In addition to MIL-STD-464, other sources of information should be considered, where applicable, that may aid in tailoring an operationally representative test EME. For systems that may encounter EMEs generated by hostile/enemy/foreign sources, there are databases that characterize these sources, too; however, most of the databases for hostile/enemy/foreign sources (as well as MIL-HDBK-235) are classified. Site surveys of areas where the system may operate can be helpful in developing a test EME as they contain measured EME data. However, the value of the data may be limited if the system operates in a dynamic and/or unpredictable battlefield environment. Finally, the system's specification may define the environment in which the system is required to operate. Testing to higher EMEs can be cost-effective for potential future applications of the system, as this could preclude the need for a future retest.

Test data from evaluations of system/system subsystems (that is, testing to show compliance to the requirements of MIL-STD-461) should also be examined in developing a test EME. The data from past testing could reveal potential areas/EME frequencies that may require investigation during evaluation of the system, as well as identifying potential trouble spots that could affect the progress of the system's evaluation.

Another factor that may influence the development of a test EME is the modes/phases of operation in which a system may function while exposed to an EME. A system may function in one mode while in one environment, but may be operating in a different mode when exposed to a different EME. A typical example is a shipboard missile in flight. During its launch phase, the missile will be exposed to the EME generated by the shipboard emitters; however, as it enters its terminal phase of flight, the missile will be exposed to an EME generated by the target's emitters. Furthermore, different electronic components of the missile will be active during its terminal phase of flight that were not active during the missile's launch phase. Therefore, two sets of test EMEs may have to be developed, one to represent the missile's launch phase and one to represent the missile's terminal phase; or the most stringent environment of both can be tested.

Testing in EMEs at every frequency and in every operational mode may provide the most rigorous characterization of a system's performance, although program resources may not allow for this thorough characterization. Therefore, tailoring of required testing is essential to take into account the EMEs and operational modes that are deemed to represent the most critical and probable operational environments/modes, and S4 phases, as well as the operational

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characteristics of the system (response bandwidths, operational frequencies, mode durations, and so forth). The required testing may be achieved with careful tailoring considerations.

5.3 Selection of a Test Facility. A facility to test the system should be chosen such that the objectives of the test can be adequately met. In choosing a test facility, the following questions should be considered:

a. Does the test laboratory have RF generating equipment capable of generating the test EME needed for the planned system evaluation?

b. Is the size of the laboratory test area(s) able to accommodate the system, and its required support equipment, to be tested?

c. What equipment is available to support the system?

d. What is the electrical power that is available to support the system and test support equipment?

e. For systems that require fuel sources (that is, aircraft, vehicles), will the fuel sources be readily available?

f. What facility restrictions exist that may impact the operation of the system (that is, safety, hazardous material (HAZMAT), and so forth)?

g. If the system/program/test data will be classified/controlled, what is the capability of the laboratory/facility to secure these items? What is the general security capability of the test laboratory and its facilities?

h. What must be done to ship the system and required support equipment to the laboratory? Is special transport required for shipping?

i. Where is (are) the laboratory(ies)/facility(ies) located?

j. What is/are the cost/fees that the test laboratory will charge to conduct testing?

k. Is the test laboratory a Government facility with test and evaluation authority? This can be cost-effective for potential future applications of the system, as this could preclude the need for a future retest.

There are generally three types of test facilities used for evaluation of systems in their external RF EMEs: open air test site (OATS), anechoic chamber, and reverberation chamber. A test laboratory may have one or more of these types of test facilities available for use. Such a test laboratory does offer some flexibility in testing; however, if more than one type of facility is being considered to support testing, the logistics of moving the SUT/test setup from one facility to another facility will have to be considered.

5.3.1 OATS Facility. An OATS facility is simply an outdoor area designated for testing. The area is generally unenclosed (except, perhaps, by security fencing) and may have a

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conductive ground plane to provide for more efficient transmissions at lower frequency EMEs and/or to provide proper equipment grounding points. The facility may have built-in power availability or may have power supplied to the facility by mobile power generators. An example of a typical ground plane facility is shown in FIGURE 1.



FIGURE 1. Typical OATS test facility with ground plane

The advantages of using an OATS are the following:

a. Generally able to accommodate large systems for testing.

b. EME reflections are minimized.

c. EMEs are generally predictable and therefore easy to characterize.

d. Minimal restrictions on EME generation based on physical constraints (that is, boundary conditions, resonant cavity effects, very low-quality factor (Q)).

e. Hazards such as fire, engine noise and exhaust (that is, from aircraft, vehicles, and power carts) are easier to mitigate than hazards in enclosed facilities.

f. Less physical restrictions to mitigate for testing RF based systems (that is, radar, global positioning system (GPS), transceivers).

g. Because testing is conducted at specific antenna locations and EME polarizations, EME susceptibility POEs are easier to identify.

The disadvantages in using an open test site are the following:

a. Exposure to weather; severe weather can add hazards and delays to testing.

b. Legal restrictions on EME generation [that is, Federal Communications Commission (FCC) regulations, National Telecommunications and Information Administration (NTIA) regulations, laboratory policies/restrictions for open-air transmission of an EME].

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c. Potential in causing unintentional EMI in equipment and systems that are not part of testing and/or belong to other activities.

d. More difficult to control ambient EME than in enclosed facilities.

e. Security concerns; the test item(s) and support equipment will be out in the open and potentially within view of unauthorized personnel and/or surveillance aircraft/spacecraft.

f. If the system is large, a test may require a large number of test positions per EME in order to sufficiently characterize the system. A large number of test positions will likely increase test time and cost, and would result in the analysis of a large amount of data.

g. Increased chance of inadvertent personnel exposure to the EME.

h. Potential ground reflections off the ground plane that could affect calibration or system exposure.

5.3.2 Anechoic Chamber. The anechoic chamber is a type of shielded enclosure lined with radar absorbent material (RAM) to reduce or eliminate RF reflections. The RAM in the enclosure comes in different shapes (usually pyramidal) and sizes designed to efficiently absorb electromagnetic (EM) energy in a certain frequency range. Multiple types/sizes of absorber material are generally used in anechoic chambers to increase the effective operational frequency range of the chamber. RAM is also rated as to how much EME power it can absorb/handle, making it imperative that the test director confirms that the anechoic chamber can handle testing at the field levels required for the system's EME evaluation. FIGURE 2 shows a typical anechoic chamber.



FIGURE 2. Typical anechoic chamber

The advantages in using an anechoic chamber are the following:

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a. No exposure to weather, as testing is conducted indoors.

b. No restrictions associated with spectrum usage.

c. No unintentional EMI in equipment and systems that are not part of testing and/or belong to other activities (that is, equipment in nearby office buildings).

d. Better control of ambient EME as testing is isolated from the external environment.

e. Eliminates exposure of non-test personnel to the test EME.

f. EME susceptibility POEs may be easier to identify based on specific antenna locations and EME polarizations.

g. Security controls may be minimized since access to the test location is better controlled.

h. EME reflections are minimized.

i. EMEs are generally predictable and, therefore, easy to characterize.

The disadvantages in using an anechoic chamber are the following:

a. Chamber dimensions may limit the size of systems that can be tested.

b. If the system is significantly large, a test may require a large number of test positions per EME in order to sufficiently characterize the system. A large number of test positions would add to test time and to cost, and would result in the analysis of a large amount of data.

c. May be restricted from testing of lower frequency EMEs due to the physical constraints of the chamber and/or absorber material.

d. Hazards such as fire, engine noise, and exhaust (that is, from aircraft, vehicles, and power carts) may require more mitigation/work-around procedures than hazards in OATS facilities. A system/equipment that contains/uses hazardous components (that is, flammable fuels, explosives, lithium batteries) may be severely restricted for use, or prohibited.

5.3.3 Reverberation Chamber. The reverberation chamber is a highly reflective (high Q) shielded enclosure that is excited by an RF source at frequencies where the chamber dimensions are large compared to excitation wavelength. The simultaneously existing modes within the chamber are stirred by a field-perturbing device referred to as a paddle-wheel tuner. As the tuner assembly is rotated by means of a stepper motor, the complex standing-wave patterns are altered significantly and the SUT is exposed to randomized fields from all directions and polarizations. Because of these reasons, the reverberation chamber is an effective tool and can be used as a supplement to traditional open-air and anechoic chamber test techniques. FIGURE 3 shows a typical reverberation chamber.

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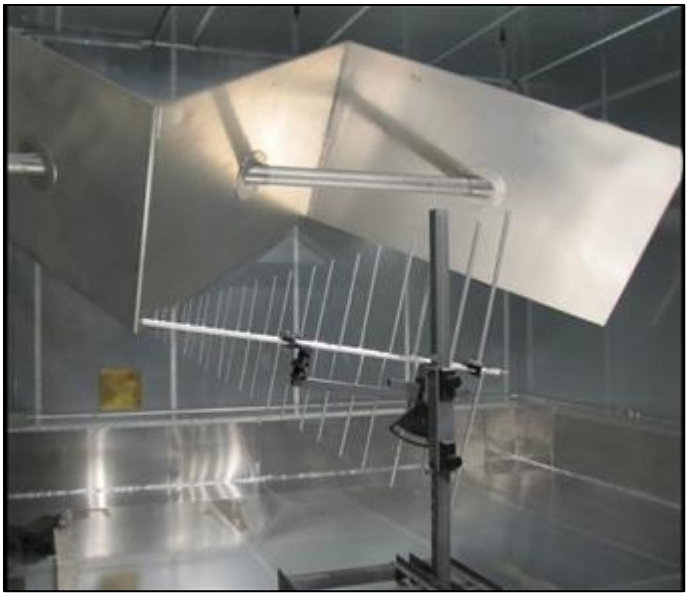


FIGURE 3. Typical reverberation chamber

The usable frequency range of a reverberation chamber, particularly the lowest usable frequency (LUF), should be understood and taken into consideration as part of the test planning process. In order to ensure that the EME inside the chamber is statistically uniform and isotropic over the course of a complete tuner rotation, the chamber must have a sufficient mode density at its LUF. The LUF can be determined by using the equation below to solve for the frequency where the number of modes (N) is equal to 60.

$$LUF = c\left(\sqrt[3]{\frac{3N}{8\pi abd}}\right) \qquad (5\text{-}1)$$

where:

a, b, and d are the chamber internal dimensions in meters
N is the Number of modes in the cavity ($N \geq 60$)
LUF is the lowest usable frequency in Hz
c is the speed of light (3×10^8 m/s)

There are two modes of operation for a reverberation chamber, defined by the way the paddle-wheel tuner is rotated: mode-stirred and mode-tuned. In a mode-stirred operation, the chamber is excited at a single frequency while the paddle-wheel tuner is continuously rotated over 360 degrees. Mode-stirred operation is often not the preferred mode of operation for external RF EME tests, since the number of samples (for example, the response intervals of the SUT) may vary significantly depending on the rotation rate of the tuners and the response time of the equipment. Furthermore, at higher frequencies, where the modal density of the chamber is higher, the rate of change for the modal structure in the chamber, which directly affects the EME levels impinging on the SUT, can change very quickly for moderate tuner rotation rates.

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In a mode-tuned operation, the paddle-wheel tuners are moved in statistically determined increments and held stationary while a measurement is made at each position. A statistically uniform and isotropic environment is achieved by indexing the tuners through a 360-degree rotation for a given input frequency and power level.

The advantages and disadvantages in using a reverberation chamber for testing are basically the same as the advantages and disadvantages in using anechoic chambers for testing, with the following exceptions:

The advantages in using a reverberation chamber are the following:

a. The high Q characteristic allows significant field levels to be generated with relatively small input power.

b. Because the SUT is exposed to randomized fields from all directions and polarizations by the rotation of the paddle wheel tuners, exposure of the entire SUT to the proper test EME reference field levels at any time during a test run is statistically assured and uniform.

The disadvantages in using a reverberation chamber are the following:

a. Because the SUT is exposed to randomized fields from all directions and polarizations by the rotation of the paddle wheel tuners, determining EME POEs into the SUT may be difficult to determine and require additional investigation, as the EME polarization and direction is obscured.

b. Chamber calibration required prior to conducting test.

c. Frequency range is limited.

d. Personnel may not be allowed to be located within the test chamber during testing.

5.4 Test Plan. After the initial particulars of a planned system evaluation have been considered (EME, selection of test site, modes/features to be evaluated, and so forth), a test plan must be created to describe, in detail, the test criteria to be implemented to evaluate the system in the EME. The test plan is developed to describe the methods to be used to evaluate the system in its operational EME. An overview of the information that should be included in a test plan is provided in the following paragraphs.

5.4.1 System Background**.** The development of test criteria and procedures is dependent on having a good understanding of the system and how it is intended to be used. Therefore, the test plan should include background information for the system that identifies who the system is designed for, what the system is designed to do, and the platforms on which the system will be used.

5.4.2 Purpose of Test. The test plan should clearly identify the reason(s) for conducting the test and the criteria to which the system is being evaluated.

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5.4.3 Scope of Test. The objectives of the test should be clearly identified in the test plan as should any limitations of the test and/or trade-offs that have been made during test planning.

5.4.4 System Test Configuration. The SUT, as well as subcomponents and any support equipment or other unique hardware planned for use during the test, should be documented in the test plan. Furthermore, details concerning the setup of the SUT at the test facility should be included as well. All applicable S4 phases and platforms should be considered, and any phase that is not to be tested should have supporting rationale.

5.4.5 Test Facility. The test plan should provide a description of the test facility and identify any unique features that may affect test setup and/or test execution.

5.4.6 Test EME. The parameters (for example, frequency, polarization, modulation, pulse width, pulse repetition frequency (PRF), and peak and average field intensity) for the test EMEs to which the system will be exposed during the external RF EME test should be defined in the test plan. The test plan should also include discussion on how the test EMEs were derived and how they correlate to the tactical EMEs that the system is expected to be exposed to during operational employment. If multiple operational phases with different test EMEs are to be evaluated, these should be clearly identified in the test plan. The test EME may also be modified to maintain maximum permissible exposure (MPE) limits if personnel will be exposed for specified operating modes.

5.4.7 Instrumentation. The method(s) used to monitor system performance during the test, whether it is via a test set or specialized telemetry system, or by direct observations by test personnel, should be documented in the test plan, and be determined early in the test planning stages. The design of any test sets or specialized telemetry systems used to collect data should be discussed in the test plan to explain how the equipment incorporates good EMI design practices such that it is likely to withstand exposure to the test EMEs and that its interface with the SUT does not introduce any RF POEs. The response time of the telemetry equipment should also be discussed to ensure it is sufficient for monitoring system performance. Special “built-in” test circuitry may be necessary to allow multiple firing events to be exercised during testing to verify proper operation of the trigger circuit.

5.4.8 Evaluation Criteria. The criteria (pass/fail) that will be used to evaluate system performance, as well as specific metrics, for all critical system functions should be agreed upon prior to the test and defined in the test plan. These metrics will be used during the test to determine if a system response constitutes a vulnerability as opposed to a susceptibility or nuisance.

5.4.9 Security Classification of Test Data. Susceptibilities and/or vulnerabilities that may be identified as a result of the system’s exposure to the test EME may be classified. Therefore, the test plan should identify how such data will be treated based on the security classification guidance (SCG) provided by the original classification authority.

5.4.10 Safety. The test plan or other related safety documentation should include discussion of potential safety hazards that may be encountered during the execution of the

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external RF EME test and provide mitigation plans/procedures that will be employed to reduce risk. These hazards may include Hazards of Electromagnetic Radiation to Ordnance (HERO), Hazards of Electromagnetic Radiation to Personnel (HERP), and Hazards of Electromagnetic Radiation to Fuel (HERF), EMI, HAZMAT, and general occupational safety and health (OSH) risks. The facility and system standard operating procedures (SOPs) should be referenced and should cover the procedures the facility uses to mitigate normal test hazards. A system-specific SOP may be required to be developed to delineate any hazards and risks associated with the SUT test operations.

5.4.11 Test Procedures. The step-by-step functional/checkout procedures that will be used to evaluate the system during the external RF EME test should be delineated in the test plan, as well as the pass/fail criteria specific to the system mission.

5.5 Test Meetings. During the course of planning for a test, there should be a series of test planning meetings/teleconferences between all pertinent parties, including test personnel, support personnel, program representatives, evaluators, and the system contractor. The meetings should be held to verify test objectives, develop test logistics (such as delivery of the SUT to/from the test laboratory), and address developing issues that could impact SUT delivery and test schedule. The frequency at which test planning meetings need to be held depends how the acquisition develops and what issues are encountered.

It is also important that test planning considerations are made for major procurement program meetings such as preliminary design reviews (PDRs) and critical design reviews (CDRs). PDRs, CDRs, and other technical meetings can provide critical insight into how the system under development is designed and is intended to function in its operational environment, and how the system operational environment is defined.

Prior to the commencement of testing, there should be a test readiness review (TRR) meeting with all key test participants. The TRR is the final test planning meeting that has the purpose of assuring the SUT program representatives and system contractor that the test team and test laboratory are ready to receive and evaluate the SUT, and of addressing/mitigating any last-minute issues that may surface.

6. CONDUCTING THE TEST

6.1 Introduction. Once the steps outlined in the previous section have been addressed and a test plan has been developed and approved, the test activity is ready to begin execution of the external RF EME test. This section discusses the different aspects of test execution, including safety considerations; generation, calibration, and application of the test EME; test setup; and methods for evaluating susceptibilities to the test EME.

6.2 Safety Considerations

6.2.1 SOP. There are several types of safety hazards inherent to the execution of external RF EME tests, which should be identified and mitigated prior to test execution. These safety hazards may vary due to the different setups and equipment used for external RF EME testing but include the following: HERO (some non-weapon electrically initiated devices (EIDs)

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may be installed on the SUT), HERP, HERF, EMI, HAZMAT, and general OSH risks. Once these hazards have been identified by the test activity, they should be documented in the SUT SOP(s) or equivalent document(s), along with mitigation procedures to reduce the risks associated with execution of external RF EME tests to a level deemed acceptable by the test activity. Prior to the start of the external RF EME tests, all test personnel should be fully familiar with the safety precautions and contents of all applicable SOP(s). The safety considerations discussed in the test plan for the hazards inherent to the SUT must also be briefed to all personnel involved in the test.

6.2.2 Safety Briefings. A safety briefing should be conducted prior to the start of each day of testing to review all safety hazards and remind all test participants of the mitigation procedures that have been implemented for the test. This daily briefing affords everyone an opportunity to gain a better understanding of the hazards and mitigation procedures, ask questions, offer suggestions to improve safety, and address issues that may arise during testing.

6.2.3 HERP. During performance of a radiated susceptibility test such as an external RF EME test, there is potential for exposing test personnel to high-level test EMEs. Personnel exposure should be limited in accordance with DoD Instruction (DoDI) 6055.11; therefore, all involved personnel should familiarize themselves with this instruction. IEEE C95.1-2345 and IEEE C95.6 are referenced as the test exposure limits in DoDI 6055.11, which is the umbrella document that specifies the general policy, responsibilities, and procedures for protecting personnel from exposure to EMEs.

In addition to direct exposure to EMEs, test personnel have to be aware of currents generated on the SUT and other exposed equipment that is coupled into the equipment from the EME fields. These contact currents, if high enough, can cause burns/tissue damage to test personnel who come direct in contact with the exposed equipment. Applicable limits for contact currents are also addressed in IEEE C95.1-2345/C95.6.

6.2.4 HERF. The SUT, as well as support equipment (that is, power carts), may require fuel to operate. Being generally flammable in nature, these fuels and their vapors could generate HERF issues, depending on the volatility of the fuel being used. NAVSEA OP 3565, Volume 1 (Chapter 6) contains a detailed discussion of HERF and provides guidance for mitigating the associated risk.

6.2.5 Personal Protective Equipment (PPE). There are several types of PPE that may be required during an external RF EME test. Each test operation should be evaluated to determine if PPE is required to provide an extra measure of protection against specific hazards. PPE also should be included in the SOP for the test operation. The list of required PPE may include hearing protection, eye protection, and/or safety footwear.

6.3 Generation of Test EME.

6.3.1 EME Generation Equipment. The proper selection of EME-generation equipment is crucial to fulfilling the external RF EME test objectives. The equipment must be able to cover the test frequencies and achieve the required reference field strengths/power densities that will satisfy testing requirements.

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The basic setup for an EME-generation system consists of two components: a signal generator (modulator, RF source, power supply) and a transmitting antenna with the appropriate cabling and/or waveguide. The signal generator generally has built-in modulation capabilities such as amplitude modulation (AM) and frequency modulation (FM); however, an additional component, the pulse generator, may be needed for pulse-modulated signals generated by gating the output from the signal generator to the required pulse widths and PRFs. FIGURE 4 shows the general setup for EME generating equipment with the inclusion of monitoring systems.

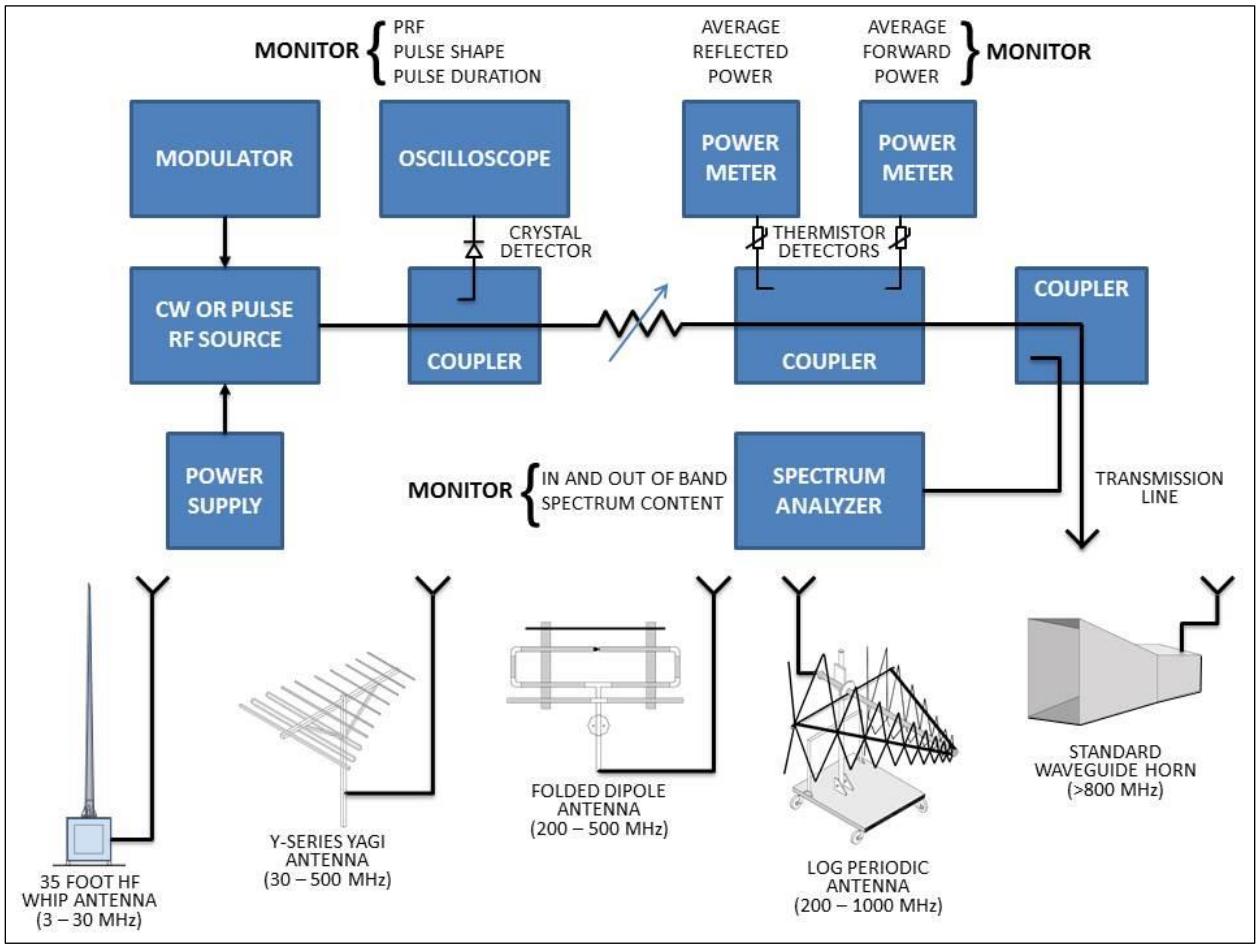


FIGURE 4. Example of EME-generation equipment setup

6.3.1.1 Signal Generators. Signal generators come in a wide variety of products and are built by a number of manufacturers. Most signal generators have a maximum output on the order of 1 to 10 milliwatts (mW). Technical selection criteria for signal generators generally revolve around the frequency coverage and modulation characteristics of the generators. There may be a need for more than one signal generator in order to completely evaluate the SUT over the required frequency bands.

6.3.1.2 Amplifiers. As with signal generators, there are different types of amplifier systems available for testing. These amplifiers are capable of transmitting EMEs over a wide range of frequencies and power outputs. Depending on the scope of the external RF EME evaluation, the test may require more than one type of amplifier. Also, some amplifiers, such as

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solid-state amplifiers, can be used relatively easily for automated swept frequency tuning susceptibility testing over a wide frequency range. Some of the more common types of amplifiers are summarized in the following paragraphs.

6.3.1.2.1 Solid-State Amplifiers. Solid-state amplifiers are amplifiers that use solid-state circuitry (that is, transistors) that are designed to step up input voltages to higher output voltages. The amplifiers are generally broadband and typically, depending on the amplifier, generate average power outputs between 100 watts (W) and 50 kilowatts (kW) providing up to a 60- to 70-decibel (dB) gain. The solid-state amplifiers generally are used to amplify input signals that are unmodulated continuous wave (CW), AM modulated, or FM modulated, and also generally have a maximum signal input of approximately 1 mW. In limited circumstances, the power amplifiers can also be used to amplify pulse-modulated input signals, but will result in a peak power output that is equivalent to the maximum-rated power output of the amplifier. For example, a pulse-modulated signal with a 10-percent duty cycle that is input into a 1-kW power amplifier will produce a maximum of up to 1 kW of peak output power and only 100 W of average power.

6.3.1.2.2 Traveling Wave Tube Amplifiers (TWTAs). TWTAs are high-power amplifiers that use a form of thermionic valve or electron tube known as a traveling wave tube (TWT). The TWT has a high bandwidth compared to many other designs, although it is possible to utilize narrowband designs when required. TWTAs may be used to generate EMEs with frequencies ranging from around 300 megahertz (MHz) up to 40 or 50 gigahertz (GHz), providing gain levels of up to 40 dB in a single device. Two types of TWTAs are generally available. Broadband TWTAs generally use helix TWTs typically capable of developing average power levels up to 2.5 kW. Narrowband TWTAs employ coupled cavity TWTs and are capable of developing average power levels up to 15 kW. TWTAs may also be used to amplify pulse-modulated input signals, but like the solid-state amplifiers will result in a peak power output that is equivalent to the maximum-rated power output of the amplifier.

Narrowband TWTAs, which use a series of coupled staggered cavities, are bandwidth-limited, typically to 10 to 20 percent of resonant frequency. Broadband TWTAs, which use a slow-wave circuit called a helix, have no particular bandwidth limitations and can practically achieve a 100-percent (octave) bandwidth. However, as indicated in the preceding paragraph, broadband TWTAs tend to have lower power outputs than narrowband TWTAs. Overall gains for both types of TWTAs typically range between 30 and 60 dB.

6.3.1.2.3 Klystron Amplifiers (KAs). A KA uses a tube called a klystron, a linear beam device where the electron flow in the tube is in a straight line focused by an axial magnetic field (H-field). The power levels of KAs are achieved through the use of large beam voltages and currents. KA efficiency can be as high as 70 percent. KAs generally operate in the ultra-high frequency (UHF) (300 MHz to 3 GHz) frequency range and with two or more cavities. The power gain of the klystron is dependent on the voltage and current as well as on the number of cavities used. The larger the number of cavities employed, the larger the gain that can be obtained. However, there is a practical limit on the number of cavities that can be used because of the onset of RF instability. For external RF EME test applications, KAs are generally used with pulse-modulated signal sources.

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6.3.1.2.4 Magnetrons. Magnetrons are generally used to generate high peak field strength EMEs with frequencies above 1 GHz. The magnetron tube is composed generally of an anode, cathode, antenna, and magnet. The cylindrical anode structure contains a number of equally spaced cavity resonators with the anode surface adjacent to the cylindrical cathode. Permanent magnets are used to provide the necessary H-field, which is perpendicular to the electric field (E-field) between the cathode and the anode. RF output is generally transferred to the EME generating antenna via a waveguide, but high-power coaxial cable is used for frequencies between 1 and 2 GHz.

Magnetrons are generally used with pulse-modulated RF sources typically with duty cycles around 0.1 percent. Magnetrons can generate high peak power levels (up to an order of 1 megawatt (MW) or more), but average power levels will be a fraction of the peak values as dictated by the EME duty cycle. Furthermore, magnetrons have operational bandwidths that are small relative to their operational frequencies so that several magnetrons may have to be used to cover the planned test frequency range. The direct current (DC)-to-RF power-conversion efficiency typically ranges from 50 to 85 percent.

6.3.1.2.5 Crossed-Field Amplifiers (CFAs). CFAs share several characteristics with magnetrons. Both contain a cylindrical cathode coaxial with an RF structure, and each of these tubes constitutes a diode in which an H-field is established perpendicular to an E-field between the cathode and the anode. Another similarity is that their RF structure serves as the electron collector and must be very rugged. The key difference is that CFAs use a delay line to introduce an RF phase delay, which thereby allows it to interact more efficiently with the electron stream. Thus, amplification occurs through most of one rotation of the electrons before the signal is extracted into an output waveguide. With this scheme, CFAs are capable of achieving very high electron tube conversion efficiencies of more than 70 percent. Additionally, the output power of CFAs is obtained with relatively low beam voltages, two to three times lower than other devices at the same power level. The gain characteristic of CFAs is a highly nonlinear one and relatively low (one to two orders of magnitude lower) compared with other electron tubes. Bandwidths of CFAs are typically 10 to 20 percent. The advantages of the CFAs are their high efficiency, small size, and relatively low-voltage operation. They are capable of average power levels from 1 kW at 10 GHz to 1 MW at 1 GHz.

6.3.1.3 Out-of-Band Noise/Harmonics. While some types of amplifiers are inherently better than others, all amplifiers have the potential to emit EM "noise," which can significantly raise the background EME noise floor over the entire frequency range of the amplifier's operation. This background noise can be present even when the amplifier/signal source, while energized and in standby mode, is not transmitting the intended test EME. When testing systems with sensitive receivers, this spurious background noise could induce erroneous responses from the SUT and make it difficult to differentiate between actual responses to the test EME and the erroneous responses to the background noise. High-power band-pass filters and narrowband antennas (designed around the test EME frequencies) may have to be incorporated to mitigate the background emissions from the amplifier. Out-of-band noise is most pronounced for tube-based amplifiers (for example, TWTAs, KAs, magnetrons, and CFAs) and, in particular, with broadband TWTAs where the out-of-band noise may be present over a relatively large frequency range. It is important to pay particular attention to the level of frequency harmonics. The SUT can respond to the harmonic and not the intentional radiated frequency.

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6.3.1.4 Antennas. Antennas come in a large variety of shapes, sizes, power-handling capabilities, and frequency-band coverage. When selecting antennas to both transmit and measure the desired test EMEs, consider the following questions:

a. Do the antennas selected for testing cover the desired frequency range?

b. Are the gains of the antennas used for EME-generation adequate enough to produce the desired EME levels at the SUT/test area?

c. Can the selected EME-generating antennas handle the output power from the amplifiers required to generate the test EME?

d. Can the EME measurement antennas handle the power coupled into the antennas from the test EME? What attenuation must be used to protect the antenna/measurement receiver system?

e. Are the antennas proportioned to meet the size of the SUT test area?

Some of the most commonly used EME-generating antennas are summarized in the following paragraphs. Note that there is not a "one-size-fits-all" antenna, and that several types of antennas may have to be used to provide adequate coverage over the entire test frequency range.

6.3.1.4.1 Dipole Antennas. A dipole antenna is a type of linearly polarized rod or wire antenna that can be, if used for RF transmission, center-fed or end-fed by a signal source. The antenna is omnidirectional in azimuth around the antenna and generally has gains around 2 decibels with reference to isotropic radiator (dBi). For a simple dipole without a tuning network, and where the antenna is designed to radiate or receive a single frequency, the antenna will generally have a length equivalent to some multiple of a wavelength; half ($\lambda/2$) and quarter ($\lambda/4$) wavelength dipoles are the most common. To broaden its operational bandwidth, a tuning circuit/matching network is connected to the antenna to vary its electrical length (which is only equal to the physical length of the antenna for the case of a simple dipole without a tuning network). Changing the electrical length of the antenna will change its resonant characteristics, resulting in the ability to tune the antenna to different operational frequencies over a wider band than a dipole without a tuning network. The practical upper frequency limit of a dipole is around 8 GHz because of the difficulty in impedance matching a tuning network to an antenna. There is no lower frequency limit, but the antennas get larger as the operational frequencies decrease (the wavelengths get longer). The typical operational bandwidth is around 10 percent.

A special case for dipole antennas is the monopole antenna, which is essentially a dipole antenna "cut in half" by a ground plane, where the lower element of the dipole is effectively eliminated. Because of physics, the monopole still behaves like a dipole in function. A significant difference is that the ground plane acts like an antenna reflector and contributes to an increase in gain over a free-space dipole antenna. Monopole antenna gains typically range between 2 and 6 dBi. To be efficient, the radius of the ground plane around the antenna should be at least the length of the active element. Figure 5 shows a 35-foot (10.7-meter)-high monopole whip antenna used for testing systems on an OATS ground plane in the high frequency (HF) (2- to 30-MHz) frequency range. This type of antenna is generally designed to handle 10- to 20-kW input powers and can

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transmit field strengths of 200+ volts per meter (V/m) at points around 10 feet (3.05 meters) away from the antenna. Most dipole antennas used in OATS facilities provide only a vertically polarized EME.

Another special case of a dipole antenna is the folded-dipole antenna. The folded dipole is created by folding the ends of a center-fed dipole antenna toward each other into a loop arrangement (FIGURE 4). Passive reflector elements are placed in front of and at the rear of the antenna to provide directivity and gain. Folded dipoles are generally designed to operate in the very high frequency (VHF) and UHF frequency ranges and typically have gains of around 5 dBi. The antennas can also be oriented to give a horizontal or vertically polarized EME.

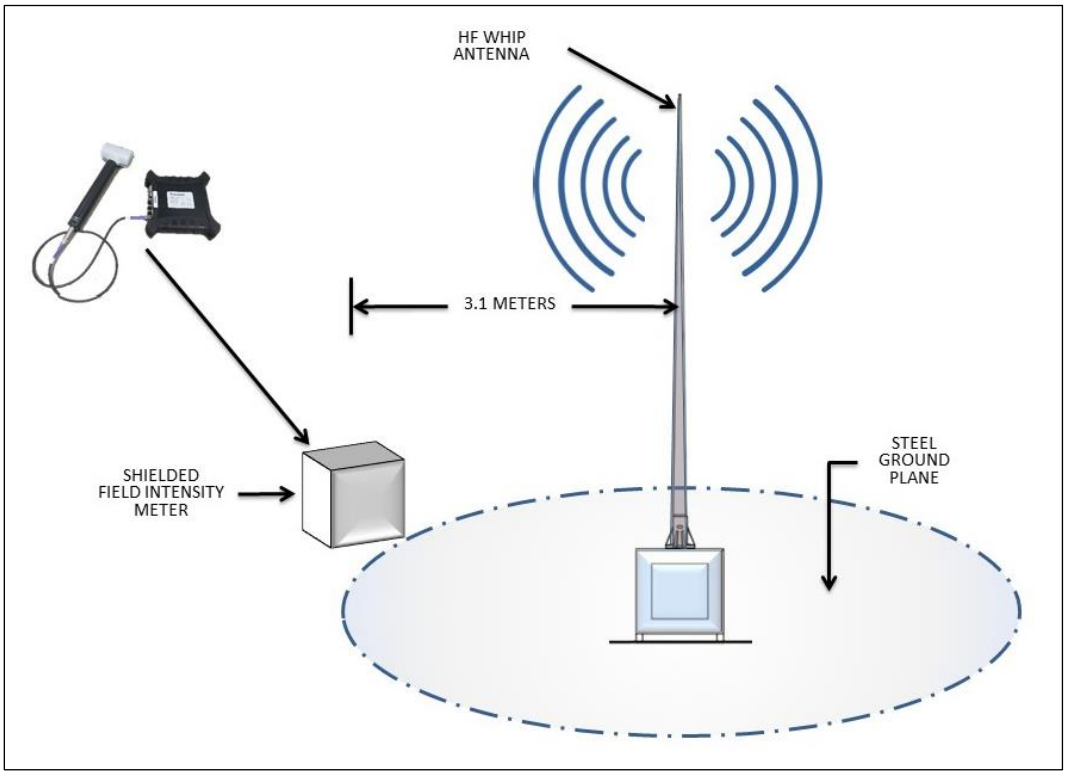


FIGURE 5. HF frequency range calibration technique

6.3.1.4.2 Yagi Antennas. A Yagi antenna is an antenna composed of an active folded-dipole element, one or more director elements, and one or more reflector elements (FIGURE 6). Generally, the more elements a Yagi has, the higher the gain, and the narrower the beamwidth. Collectively, Yagi antenna designs generally span the 50-MHz through 2-GHz frequency range. Gains typically range between 5 and 15 dBi, with bandwidths around 5 percent. The antennas can also be oriented to give a horizontal or vertically polarized EME.

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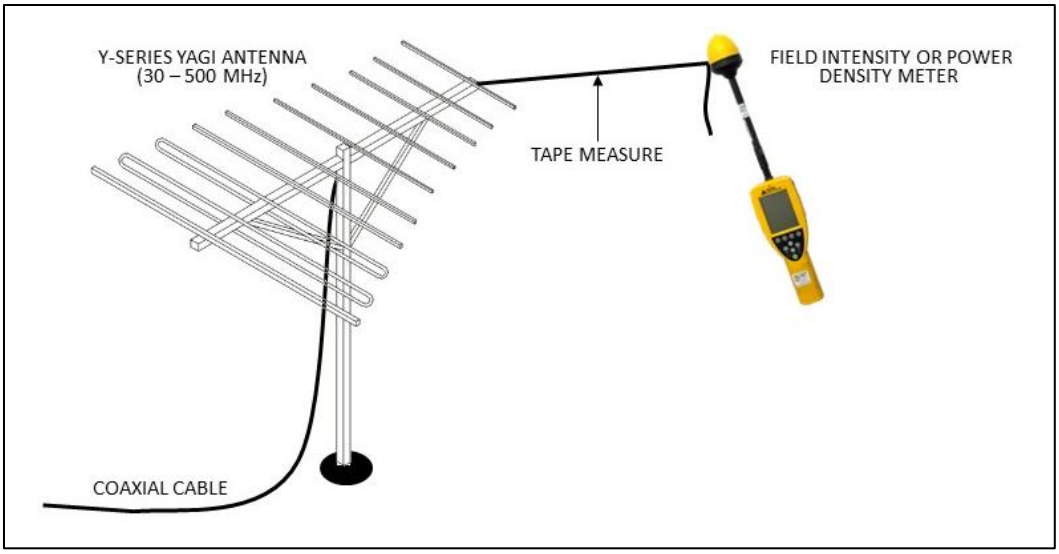


FIGURE 6. VHF frequency range calibration technique

6.3.1.4.3 Log-Periodic Antennas. The log-periodic antenna is based on the dipole element. The antenna is comprised of a set of dipoles or toothed arrays that are all active and vary in size from the smallest dipole/toothed array at the front to the largest dipole/toothed array at the rear. Usually, this antenna is constructed so the antenna terminals are located at the front (on the shortest dipole). FIGURE 4 shows a typical toothed arrayed log-periodic antenna. The key features of this antenna are its broadband nature and its relatively high front-to-back gain ratio. Collectively, log-periodic antenna designs generally span the 3-MHz through 18-GHz frequency range. Gains typically range between 6 and 8 dBi, with bandwidths around a 10:1 ratio. The antennas can also be oriented to give a horizontal or vertically polarized EME.

6.3.1.4.4 Pyramidal Horn Antennas. A pyramidal standard gain horn antenna is a type of a horn antenna with the horn in the shape of a four-sided pyramid and with a rectangular cross-section. Horn antennas are generally used to generate, or measure, EMEs with frequencies between 50 MHz and 40 GHz. The pyramidal horn antennas generally have gains between 5 and 20 dB and are narrowband with a 2:1 bandwidth ratio. High-power standard gain horns used for transmitting EMEs are predominately waveguide-fed for EMEs with frequencies above 2 GHz, and coaxial-cable-fed for EMEs with frequencies less than 2 GHz. Measurement antennas, for frequencies above 2 GHz, can be found using either coaxial RF cabling or waveguide, but predominately use coaxial RF cables for frequencies below 2 GHz. The size of a pyramidal horn antenna is inversely proportional to the frequency range in which it is intended to operate; consequentially, standard gain horns designed to operate in the VHF and the lower (< 500 MHz) UHF frequency ranges can get very large and are generally used where large areas of EME illumination are required and with large-sized SUTs (that is, aircraft). FIGURE 4 shows a typical standard waveguide horn antenna.

A special type of horn antenna is a ridged pyramidal horn antenna. This antenna has ridges or fins attached to the inside of the horn, extending down the center of the sides. The fins lower the cutoff frequency, increasing the antenna's bandwidth with a 4:1 bandwidth being typical.

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6.3.2 EME Calibration for OATS and Anechoic Chamber Test Facilities. The test EME should be measured and verified for each frequency band or antenna to ensure proper field intensity/power density levels illuminate the SUT. FIGURE 5 through FIGURE 7 illustrate typical field calibration/measurement setups that are used in different EME frequency ranges. These field calibration/measurement setups are accomplished prior to introducing the SUT in the field. Once completed, the SUT is exposed to the field at the calibrated distance and EME level. As can be seen in FIGURE 5, the EME at HF is calibrated at a fixed distance whereas all other frequencies are calibrated for a fixed EME level. In the case of HF calibration, the field measurement sensor is placed on the ground plane at a specific distance while the source output is varied to attain the required EME level. For all other frequencies, both the distance to and power output of the source are varied until the required EME level is attained. As noted in FIGURE 5, the specific distance is defined and maintained during testing by a measuring device. Other references, such as MIL-STD-461, provide additional insight into EME verification.

NOTE: If an SUT is mounted to a structure or vehicle, it may be determined that the structure or vehicle intersects the beam radiated from an antenna prior to reaching the SUT. This point of intersection to the platform's nearest conductive surface is referred to as "first skin." In test arrangements with a first skin, the separation distance is measured from the antenna to the first skin rather than from the antenna to the SUT. If testing is conducted at the platform level, the height of the ordnance as installed on the platform must be taken into consideration. The test EME will be measured and calibrated to the nearest conductive surface ("first skin") prior to testing at each frequency to ensure proper field strength levels.



FIGURE 7. Radar (GHz) frequency range calibration technique

When the EME field intensity/power density is to be measured, the measurement antenna or field probe ideally should be placed as far as possible from the source antenna as practical. While the far field is preferred, it may not be practical due to amplifier and source antenna limitations.

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Care should be taken to be far enough away to get the item under test within the 3-dB beamwidth of the source antenna.

$$R_{ff} = \frac{2D^2}{\lambda} \text{ or } R_{ff} = 3\lambda, \text{ whichever is greater} \quad (6\text{-}1)$$

where:

R_{ff} is the far field distance in meters
D is the largest physical dimension of the antenna in meters
λ is the EME wavelength in meters

Equation 6-1 should give a reasonable approximation for R_{ff} expressed in meters. For horn antennas, D has been traditionally taken as either the largest dimension defining the horn aperture (height or width), or the length of the diagonal bisecting the horn aperture (using the diagonal to determine the antenna far field is a more conservative approximation of the far-field distance).

While measuring the reference field levels and testing at a distance equal to or greater than the far-field distance is ideal, actually attempting to achieve the desired test field levels at the transmitting antenna far field may be impractical or even impossible. For example, using the preceding equations, the far field for a 10-meter HF whip transmitting at a frequency of 30 MHz (where $R_{ff} = 900/f_{MHz}$ applies based on the antenna parameters) is 30 meters. Therefore, to generate a typical 200-V/m test EME at 30 meters (assuming a nominal antenna gain of 3 dB for an HF monopole antenna), the transmitter would have to generate an output power of around 600 kW. Most HF transmitters today generally can generate output powers of around 1 kW or less, with specialized high-power HF transmitters (such as those used at RF EME test facilities) being capable of achieving output powers between 15 and 20 kW, still significantly below the 600 kW required to generate a 200-V/m 30-MHz field at 30 meters distance. In addition, if an HF transmitter capable of a 600-kW power output existed and was being considered for use in testing, other concerns, such as MPE limits and test area location, may preclude the use of such a transmitter. Therefore, testing may have to be performed in the antenna near field at a distance where the desired field level can be obtained, or testing may have to be conducted at a field level that is less than the required EME field level if being located in the far field is desired. These trade-offs may have to be considered when testing in particular EMEs is required.

Antennas to be used for measuring EMEs should be linear antennas such as monopole/dipole antennas, biconical antennas, ridged pyramidal horn antennas, and standard gain pyramidal horn antennas. In contrast, log-periodic, Yagi, and conical log spiral antennas should not be used to measure EMEs. Log-periodic and Yagi antennas have issues concerning field variations across the antenna elements/apertures, as well as issues concerning the far field/near field, that make it difficult to reproduce measurements. In the case of conical log spiral antennas, the antenna is circular polarized, thus creating confusion as to how the antenna can be used to measure EMEs that are generally linearly polarized, and whether a 3-dB correction factor is necessary to account for the linear polarization of the EME. In addition, at certain frequencies, the antenna pattern of a conical log spiral antenna is not centered on the antenna axis.

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6.3.3 EME Calibration for Reverberation Chamber Test Facilities. The calibration methods for an EME in a reverberation chamber are somewhat different from the calibration method described for OATS and anechoic chamber test facilities in 6.3.2. Since multiple procedures have been adopted for calibrating an EME in a reverberation chamber, the procedures are not detailed in this document. The preferred method for calibration is contained in International Electrotechnical Commission (IEC) 61000-4-21, while MIL-STD-461 provides an alternate calibration method.

6.3.4 Modulated E-Field Calibration Techniques. The following paragraphs describe the techniques for calibrating modulated E-fields.

6.3.4.1 CW, FM, and AM E-Field Calibration.

a. Place an E-field probe or receive antenna in the location where the test item will be located (distance from antenna and at the height of the test item). The test item is not in the field (free-space calibration).

b. Bring up the RF in CW using the signal generator/amplifier until the specified average E-field is met.

c. Record the level of the signal generator and amplifier (decibels with reference to 1 mW (dBm) level and W).

NOTE: If AM testing is to be performed, ensure that when the required AM modulation index is applied, the peak power level will not exceed the amplifier's power capability (signal is not distorted by "clipping" the waveform).

6.3.4.2 PM E-Field Calibration.

a. Place an E-field probe or receive antenna in the location (distance from antenna and at the height of the test item) where the test item will be located. The test item is not in the field (free-space calibration).

b. Bring up the RF signal generator/amplifier until the specified average E-field is met.

(1) If using a CW amplifier (for example, solid-state, TWT, and so forth), calibrate with CW (and then the modulation will be applied during the actual test).

(2) If using a peak power amplifier, bring up the RF to the W needed to obtain the pulsed field levels.

c. Note the level of the signal generator and amplifier (dBm level and W).

6.3.5 Application of Test EME. When testing in OATS and anechoic chamber test facilities, the SUT will generally be illuminated by directional antennas placed at various points around the SUT. The points, or aspects, of illumination are generally based on critical apertures, cable runs, connectors, or subsystems that are located at these aspects. The number of aspects needed to satisfactorily illuminate the SUT can be dependent on the size of the SUT, the number

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and location of critical areas in/on the SUT, and the illumination beamwidth of the EME transmitting antenna. As a rule of thumb, the higher the frequency of the radiating EME, the smaller the beamwidth and effective aperture of the radiating antenna, resulting in more aspects at which the antenna has to be placed in order to provide satisfactory coverage of the SUT. In addition, generally, for frequencies above 30 MHz, the SUT aspects should be illuminated at two EME polarizations (vertical and horizontal) to provide a more complete picture of EME coupling into the SUT. (For frequencies below 30 MHz, the EME is traditionally only vertically polarized because of the difficulty of orienting test equipment relative to the SUT for horizontal measurements.) Also, the EME antenna should be placed at a distance far enough from the SUT to minimize near-field effects. Preferably, the distance between the antenna and the illumination area should be at the far-field distance; however, as with calibration measurements, limitations may preclude being able to set the SUT in the far field, requiring the test engineer to consider practical trade-offs to acquire valid test data. Note that the EME antenna should not be placed too close to the illumination surface. If the antenna is placed too close to an SUT surface, the SUT and antenna may form a capacitive alternating current (AC) circuit, in which some of the EME energy will manifest itself as a displacement current in the “circuit” and not as a radiated field, resulting in potential skewing of the test data/results. An indication of being too close to an SUT is a significant increase of reverse power at the transmitter. For a rotary wing aircraft, the blades must be rotated such that they are not close to the transmitting antenna. This distance needs to be taken into consideration when calibrating the test E-field since the field probe/receiving antenna needs to be at the location of the SUT to ensure adequate EME.

When testing in a reverberation chamber, EME polarizations and illumination aspects are not as critical to consider as when testing in OATS and anechoic chamber test facilities. During testing in a reverberation chamber, the SUT is effectively “immersed” in the EME, being illuminated at all aspects and polarizations. Placement of the SUT can be relatively anywhere in the chamber, provided the test setup is within the working volume and meets configuration requirements. However, because of the way the SUT is illuminated in the chamber, the polarization of an EME that causes SUT susceptibility, along with the EME POE, may be difficult to determine and require additional investigation. If EME polarization and POEs need to be known, then testing may have to be repeated at the frequency(ies) of interest in an OATS or anechoic chamber test facility using directional antennas.

Finally, there is a risk that the SUT may incur permanent damage as a result of the system's exposure to the test EME. Therefore, it is recommended to take into account features of the system's design, the full reference test field strength for each test EME, and any previous test data (if available) to assess this risk prior to exposing the SUT to the test EME. For test EMEs where the risk is not deemed to be low or acceptable, the risk may be mitigated by reducing the EME field strength level to which the SUT is initially exposed. Then, once the test cycle is completed at the lower level, the EME field strength can be increased during subsequent test cycles provided that no susceptibilities are observed to indicate the likelihood of the SUT being damaged as result of exposure to a higher EME level. Often, the EME field strength level is increased by 3 dB for each subsequent test cycle; however, other increments may be used at the discretion of the planning team. This process should be repeated until the full reference test field strength is reached, or until observed susceptibilities indicate that going to a higher field level may cause permanent damage or degradation to an SUT. While not mandatory, the preceding process will help establish a threshold of susceptibility (TOS), avoid damaging the SUT, and

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avoid test delays from having to repair or replace a damaged SUT. This includes not only the ordnance item under test, but other equipment on the platform that may be susceptible to damage.

6.3.5.1 Testing.

a. For CW, bring up the RF to the CW calibration level and run the test.

b. For FM, bring up the RF to the CW level and turn on the modulation and run the test.

c. For AM, reduce the calibrated CW signal level by the AM constant peak conservation index (CPCI) in equation 6-2 below and turn on the modulation. This reduction maintains the constant peak signal relationship between CW and AM.

$$AM\ CPCI\ (dB) = 10\ log_{10}\left(\frac{1}{1+m}\right)^2 \qquad (6\text{-}2)$$

where:

m is the modulation index (0.8 for 80-percent modulation depth)

d. For PM testing:

(1) If using a CW amplifier (for example, solid state, TWT, and so forth), turn on the modulation and then bring up the RF to the CW dBm level.

(2) If using a peak power amplifier, bring up the RF to the W noted during calibration.

6.3.5.2 Thresholding. When an anomaly is noted during testing, a threshold will be required. See 6.5 for additional information.

a. CW: Bring up the RF in CW in small increments for each test run until the anomaly is noted. Calculate the field from that power level using equation 6-3.

b. AM: Bring up the RF in the AM in small increments for each test run until the anomaly is noted. Turn off the modulation and adjust up using the *AM CPCI* and note the power level. Calculate the field from the unmodulated power level using equation 6-3.

c. FM: Bring up the RF in the FM in small increments for each test run until the anomaly is noted. Turn off the modulation and note the power level. Calculate the field from the unmodulated power level using equation 6-3.

d. PM:

(1) For CW amplifiers (for example, solid-state, TWT, and so forth), bring up the RF in the PM in small increments (for example, 1/16, 1/8, and so forth) for each test

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run until the anomaly is noted. Turn off the modulation and note the power level. Calculate the field from the unmodulated power level using equation 6-3. (NOTE: For PM testing, before removing the PM and going to CW, consideration must be made for the fact that the average test level will go up and the test item may need to be removed first.)

(2) For peak power amplifiers, bring up the RF in small increments until the anomaly is noted. Note the power level and calculate the field from that power level using equation 6-3.

$$E_T = \sqrt{\frac{P_T}{P_C}} * E_C \qquad (6\text{-}3)$$

where:
E_T is the threshold E-field
E_C is the calibrated E-field
P_T is the power obtained at threshold (when the modulation is turned off) when the anomaly occurred
P_C is the power of the calibrated e-field (when the modulation is turned off)

6.4 Test Setup. The SUT should ideally be set up in the configuration in which it would be when exposed to the operational EME. However, because test setup limitations will probably be encountered, setup trade-offs may have to be considered, implemented, and balanced in such a way that, while the SUT somewhat deviates from its actual tactical configuration, the evaluation will still remain valid and realistic. Furthermore, SUTs that have several modes of operation might have to be tested using a few key selected/worst-case modes, since testing the SUT in all of its operational modes potentially could be prohibitively time-consuming. Test support equipment interfaced with the SUT (that is, power supplies, monitoring/control equipment, sensors, and so forth) that is not normally part of the SUT should be isolated in such a way that the support equipment does not create an EME coupling path that results in erroneous test data. FIGURE 8 shows illustrations of test setups in the anechoic chamber and reverberation chamber test facilities.

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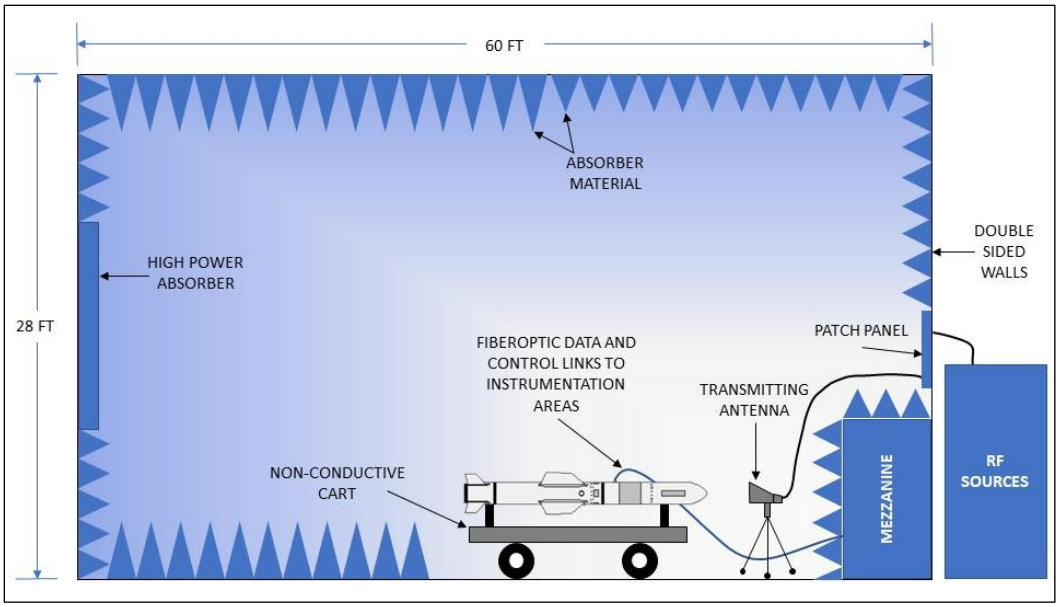


FIGURE 8. Block diagram example of a typical test setup in an anechoic chamber

For more detailed descriptions and explanations of test setups, review the latest versions of MIL-STD-461, Test Operations Procedure (TOP) 01-2-511, or Allied Environmental Conditions and Test Procedures (AECTP)-500.

6.5 Susceptibility Investigation. A system is said to have a susceptibility if any form of response is recorded as a result of exposure to a level of electromagnetic radiation (EMR) representative of the system's intended EME. A vulnerability is considered a special subset of a susceptibility; specifically, a system is said to have a vulnerability if the system exhibits a definite degradation as a result of having been subjected to a level of EMR representative of the system's intended EME. Therefore, the TOS is the E-field level (within 3 dB) at which the system functions operationally without any form of susceptibility or response to the EME, and the threshold of vulnerability (TOV) is defined as the E-field level (within 3 dB) at which the system response is no longer a vulnerability to the mission (for example, the EMI no longer is preventing the system from performing its intended mission).

When conducting a test, the test director may encounter anomalies in the operation of the SUT that have to be investigated. If an anomaly is encountered during exposure of the SUT to the test EME, the first question that must be asked is whether the anomaly is actually being caused by the test EME coupling into the system, or if the cause was independent of the system's exposure to the test EME. This question generally can be answered by simply re-exposing the SUT to the EME and observing if the anomaly can be repeated. If it is repeatable, then the anomaly can be regarded as a susceptibility to the EME.

The next step, if the anomaly is identified as a susceptibility, is to determine if the susceptibility is severe enough to warrant further investigation. For instance, if the susceptibility is determined to be initially a nuisance problem (such as minor interference on a video display that does not render the display unusable) and remains a nuisance problem even when the SUT is exposed to the EME at full criteria field levels, then, while it should be noted, no further investigation would

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be necessary. However, if the susceptibility becomes significant enough to require operator intervention, or if it is determined to be a vulnerability (or becomes a vulnerability when the EME field level is increased to a level at or below the criteria field level), then it will need to be investigated to determine its impact. This impact should be assessed as part of the report.

When the observed susceptibility (or vulnerability) is determined to be significant, the test director should consider the following:

a. What is the POE for the EME?

b. Is the susceptibility dependent on EME polarization?

c. Is the susceptibility dependent on EME modulation?

d. If the EME field level is lowered, what is the threshold where the susceptibility/vulnerability disappears?

e. Is the primary mechanism of EME coupling through a cable, through an electrical switch (acting as an antenna), or through an aperture in the SUT and directly onto the SUT's electronic components?

When a susceptibility is observed and determined to be significant, the TOS and TOV should be determined. Note that the TOS may not be the same as the TOV, and that in some instances an observed susceptibility may not result in a vulnerability if the power is raised from the TOS up to the criteria field level without experiencing further degradation in system performance. When the susceptibility is observed and considered to be significant, the EME field level should be lowered (typically in 3-dB steps) until the susceptibility disappears to establish the TOS level. The field level should then be raised 3 dB to see if the susceptibility reappears and thus verify the TOS. The field level should then be raised 3 dB to see if the susceptibility reappears and thus verify the TOS. This same procedure can be used to determine the TOV; however, care should be taken when establishing a TOV as the higher field levels above the TOS could result in permanent degradation or damage to the SUT. When their respective values are established, the TOS and/or TOV can be compared to the expected operational EME field levels from emitters in the area(s) where the SUT will be installed/used to determine if these emitters have the potential of affecting SUT operation in the field. The TOS and TOV can also be used to calculate the standoff distances from potentially problematic emitters; for example, the minimum distance at which the SUT has to be located, relative to an emitter, where the emitter field levels will not detrimentally affect the SUT.

For mechanisms of EME coupling suspected to be through SUT cabling, the suspected cable or cables should first be inspected for obvious signs of cable deficiencies ("pigtail" grounds, cable shield discontinuities, absence of cable shielding, and improper connector grounding/bonding). After inspecting the cabling, as part of the investigation, consider implementing temporary engineering corrections, or "fixes," to the cabling to determine if the impact of the EME can be reduced or eliminated. Temporary fixes may include wrapping the cables/connectors in conductive metal tape (usually copper or aluminum), feeding the cable(s) through a metal braid sleeve or conductive conduit, or attaching ferrites to the cabling. These temporary fixes can help

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determine if, and to what degree, more permanent engineering solutions (for example, filtering and/or shielding) will be necessary to mitigate the susceptibility.

For mechanisms of EME coupling suspected to be through SUT apertures (such as holes, seams, displays, and vents), the suspected apertures may be temporarily covered with metallic shielding material applied to the apertures to determine if the impact of the EME can be reduced or eliminated. If there is more than one suspected aperture, all apertures should be covered initially to eliminate the EME, and then each aperture should be uncovered one by one until the susceptibility reappears. The uncovered aperture should then be re-covered, and then any remaining apertures should be uncovered, one by one, to determine if more than one aperture contributes to the susceptibility. The temporary shielding that can be used includes conductive tape/foil and/or screen material (especially used to cover displays). As with the temporary engineering corrections implemented for cables, the temporary aperture fixes can help determine if, and to what degree, more permanent engineering solutions (for example, welded seams, display screens/conductive display glass), will be necessary to mitigate the susceptibility.

6.6 Supplemental Test Techniques. System-level testing at the full external threat environment has historically been the primary method employed when conducting external RF EME evaluations. However, due to increasing EMEs, this may not always be feasible. An alternative method is to decompose the testing into multiple lower power tests and then statistically determine the probability-of-effect. The distribution of the maximum fields within the system enclosure is first measured using complex cavity test techniques. These techniques can be scaled to the external threat environment by the frequency-dependent SE transfer function of the system enclosure. Then, the internal equipment threshold can be directly measured through susceptibility testing (at levels representative of internal fields). Finally, the probability-of-effect is determined from the system hardness margin (for example, the ratio of equipment threshold to the maximum environment).

Additional supplementary techniques, such as conducted susceptibility (CS) testing, can be used when the transfer function between radiated and conducted environments is known. For example, the MIL-STD-461 conducted susceptibility bulk cable injection test (CS114) can be used as an alternative to exposing cables to radiated fields in the range of 10 kHz to 200 MHz.

7. DATA ANALYSIS

7.1 Introduction. Testing systems for susceptibility to external RF EMEs is a risk-reduction technique to be culminated in a test report. The data analysis will result in the justifiable conclusion and recommendations for safety and reliability. Systems exposed to more than one of the defined EMEs must use the worst-case composite of the applicable EMEs. External RF EME refers to compatibility with, but is not limited to, EMEs from like platforms (such as aircraft in formation flying, a ship with escort ships, and shelter-to-shelter in ground systems) and friendly emitters. Compliance must be verified by system-, subsystem-, and equipment-level tests, analysis, or a combination thereof. Testing systems to external RF EMEs must be culminated by a test report with a safety and reliability assessment. The external RF EME test report requires several fundamental components to be effective in communicating perceived risk and the recommendations for risk mitigation, based on the test results. The test report includes conclusions and recommendations based on the test results and applicability of

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those results to the mission. When susceptibility is observed, a discussion of any remedial actions and/or proposed corrective measures should be included in the report. Equally important is that the report should be written in a manner such that the results could be repeated at a later date. Therefore, the test report should include a thorough description of test configuration, photographs, or diagrams of the test setup, details of test conditions, operational procedures, test environment (including modulations), and a summary of results including TOS/TOV. Deviations from the test plan must also be documented. Note that including specific data may make the report classified.

7.2 Analysis of Detected Susceptibility. It is likely that an external RF EME test will identify susceptibilities, which require further analysis. Though the cause and effect of observed susceptibilities may not be apparent upon completion of the external RF EME test, all resources should be consulted to gather as much information possible to complete a thorough analysis of the test data.

The degree to which a system is susceptible to a given stimulus can vary depending on how the EME is coupled to a device, how much energy is coupled, properties (modulation, polarization, power) of the impinging EM waves, and how the device reacts to the stimulus. Each unique combination of these factors can suppress or exacerbate the degree of susceptibility. Susceptibility can be unique in response characteristics such as duration, simultaneous responses, bandwidth of susceptibility, and POE. Therefore, analysis of the results to develop recommendations requires consideration of all available data regarding the detected susceptibilities. Though detected susceptibilities can be unique, the type of degradation observed can be categorized into one or more of the following categories:

Minor Degradation (Minor Mission Impact):
During active exposure to EM energy, a system displays observable degradation, which does not affect, or has minimal impact to, its ability to execute at its full mission capability.

Temporary Degradation (Moderate-to-Severe Mission Impact):
During active exposure to EM energy, a system displays functional degradation, which affects its ability to execute its full mission capability. The device can return to its full mission capability after exposure ceases and may require remedial action such as system reset or power cycling to return to full system capability.

Permanent Degradation (Moderate-to-Severe Mission Impact):
During and after active exposure to EM energy, a system displays functional degradation, which affects its ability to execute its full mission capability. The device cannot return to a full mission capability after removal of the EM energy and other remedial actions are required, such as system reset or power cycling.

A subcategory of permanent degradation is catastrophic failure, which is considered the following:

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Catastrophic Failure (Severe Mission Impact):
A system that has suffered physical damage, electrical damage, or both, that renders it non-functional after removal of the EM energy and other remedial actions are taken such as system reset or power cycling.

An assessment of probability of susceptibility, degradation category of the associated susceptibility, and the severity of mission impact of a detected susceptibility should accompany the test results. The severity of mission impact, associated with the preceding categories, can be minor, moderate, or severe. Since the severity of impact is subjective, input from subject matter experts, the system/component manufacturer, and/or end user(s) may be beneficial. The probability of susceptibility assessment should consider the expected operational scenarios through all phases and modes of the system's operation. To determine probability of susceptibility, it may be beneficial to have a timeline of the system's operational modes and corresponding EMEs which could be expected.

7.3 Analysis of Risk. No data set is likely to be complete and exhaustive due to cost and schedule constraints associated with conducting testing. Therefore, risk of external RF EME is always present and will be due to known and unknown consequences of the test. Known risks are factors such as gaps in the test frequencies, illumination incompleteness, or other risks determined prior to testing. The known risks can often be quantified through statistical means prior to test commencement. Unknown risks can be those such as damage to SUT, test EME generation equipment availability, lack of sufficient test time, susceptibility bandwidth, or other unknown risks determined during or after testing. Because external RF EME testing is performed to reduce risk of susceptibility, reduction of unevaluated risks should be managed upfront to the greatest extent possible. Because unevaluated risks are unknowns, assessing risk associated with them is difficult.

7.3.1 Known Risks. Known risks are known in advance and will not change. These risks can be quantified using statistical analysis of independent variables such as aspect angle, frequency distribution, and likelihood of encounter of particular EMEs of concern. A well-planned test will minimize the known risks, but they cannot be eliminated completely.

7.3.2 Unknown Risks. For unknown risks, analysis can be used to quantify the likelihood of susceptibility in a similar manner to the way planned risks can be quantified. Engineering considerations of risk can be made based on the data collected. For example, if a particular frequency, which could not be evaluated during the test, is in close proximity to one which was evaluated, similar results could be expected. Additionally, arguments can be made that electrical length of equipment and cables can influence coupling of EMEs and, therefore, increase/decrease risk to particular sets of wavelengths. Arguments for low, medium, or high risk of susceptibility can be made dependent on the assessment of filtering, aperture size, shielding, or physical separation. However, these determinations can be subjective, and the associated risk should be determined with all factors considered.

7.4 Test Reporting. The external RF EME test report requires several fundamental components to be effective in communicating perceived risk and the recommendations for risk mitigation based on the test results. The test report should include conclusions and recommendations based on the test results and applicability of those results. When susceptibility

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is observed, or the potential for risk is high, a discussion of any remedial actions and/or proposed corrective measures should be included in the report. Equally important is that the report should be written in a manner such that the results could be repeated at a later date. Therefore, the test report should include a thorough description of test configuration, photographs or diagrams of the test setup, details of test conditions, operational procedures, test environment, and a summary of results including TOS/TOV. Deviations from the test plan must also be documented. Note that including TOS/TOV data may make the report classified.

Because the data analysis and reporting phase is the culmination of the entire external RF EME effort, it is the most important. The test report should document and capture test configuration, conditions, operational procedures, parameters, pictures of test setup, summary of data taken including TOS, discussion of any remedial actions initiated, and discussion of observed results, all in a brief narrative form. The test report must include conclusions and recommendations to provide the acquisition program and/or designated technical authority with an assessment of how the system is expected to operate during exposure to the tactical environment.

The first step of the reporting process is to perform an analysis of the test results to determine the impact of exposure to external RF EME during operational use. The majority of the recommendations and conclusions should be developed from this analysis and, therefore, the analysis is one of the most important aspects of the test effort. Analysis of the test data should be used to provide quantitative and/or qualitative assessments of risk of external RF EME based on detected responses and risks. Finally, the analysis will correlate the risk associated with these responses to the impact they may have on the system's mission during all applicable stages of its operational lifetime and associated configurations.

7.5 Conclusions and Recommendations. The test data analyses relate the susceptibilities observed to the system's associated mission impact. The analysis results lead to conclusions, which provide the basis for the recommendations section. Conclusions should include where/how EM energy entered the system, where and when susceptibility may occur, likelihood of experiencing susceptibility, and anticipated success of the system to perform in its intended operational EME. It is often unknown if observed susceptibilities and the associated risk are acceptable for the program or the user prior to the report being released. Therefore, recommendation of a solution for all observed susceptibilities should be provided and, if possible, alternative options should also be provided to allow for additional flexibility.

If the conclusion is that the probability of susceptibility is low (see MIL-STD-882 for assessing risk), a possible recommendation could be presented that specifies no modification should be made to the system or system's operational procedures. Alternatively, a recommendation to modify the system's operational procedures, to modify the system's configuration, or to accept the risk may be required for moderate-to-severe mission impacts. Recommendations should address how to effectively isolate expected operational EMEs from the affected electronics within the system. Options that may be recommended to mitigate EMI include shielding cables, shielding apertures, installing filters, installing EMI gaskets, and so forth. Other recommendations associated with the how/when/where the system is used can be just as effective as material solutions.

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8. SYSTEM CHANGES THAT WOULD SIGNAL A RETEST

External RF EME test results are directly related to the test setup and configuration of test hardware available for test. Therefore, any changes to the system design and/or installation procedures should be reviewed to determine if the external RF EME test results are applicable to the new design and/or operational configuration. The following is an overview of some design changes that may warrant supplemental external RF EME testing or a complete retest:

8.1 System Design Changes that Require an External RF EME Test.

a. Changes to cable designs such as lengths, routing, and shielding.

b. Changes to surface treatments (for example, corrosion protection) at mating seams.

c. Changes to EMR absorber, coatings or gaskets.

d. Any added aperture or antenna.

e. Changes to electronics (including electronics with the same form, fit, and function), particularly receivers and control circuits.

8.2 Operational Configuration Changes that Require an External RF EME Test.

a. Integration on a new platform (or the same platform that has been modified).

b. Integration with a new or modified launcher or fire control set.

c. Deployment in a new operational EME.

d. Changes to platform interface.

9. DOCUMENTATION

Test documentation incorporates the total assimilation of a test program. Without proper documentation, a test program could be forfeited. The test plan lays the groundwork for accomplishing the mission of certification/compliance. All the minute details for conducting the test should be captured here. The substance of the test program resides in adherence to the test procedures during the test and in the data collection process. Faithful execution of the test procedures and explicit data collection will bring the test goals into focus and lead to a successful completion of the test mission. The test report unifies the goals and objectives by summing up the test data into viable conclusions and recommendations, thus finalizing the process. The documentation for test reporting is detailed in 7.4.

10. NOTES

10.1 Intended Use. This handbook can be used by any facility engaged in E3 ordnance testing for any of the military departments.

10.2 Supersession Data. Not applicable.

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10.3 Cross-Reference of Classification and Substitutability Data. Not applicable.

10.4 Subject Term (Keyword) Listing.

E3
Electromagnetic (EM)
Electromagnetic Compatibility (EMC)
Electromagnetic Environmental Effects (E3)
Electromagnetic Radiation Hazards
Radio Frequency Electromagnetic Environment (RF EME)

10.5 Identification of Changes. Not applicable.

10.6 Technical Points of Contact. Requests for additional information or assistance on this standard can be obtained from the following:

Army

US Army Test and Evaluation Command (HQ, ATEC)
6617 Aberdeen Blvd., Building 2202
ATTN: CSTE-OPS
Aberdeen Proving Ground, MD 21005-5001
(443) 861-9664
usarmy.apg.atec.mbx.atec-hq-g3-5@mail.mil

Army Test and Evaluation Command (ATEC)
E3 Test Division (E3 for Missile System Functions)
ATTN: TERT-ECE
Building 8978
Redstone Arsenal, AL 35898-8052
(256) 313-6970
usarmy.redstone.atec.list.rtc-jocg@mail.mil

Army Test and Evaluation Command (ATEC)
White Sands Test Center
Survivability, Vulnerability and Assessment Directorate
ATTN: TEWS-SVE (ATTN: External RF EME)
21225 Headquarters Avenue
White Sands Missile Range, NM 88002
(575) 678-6307
usarmy.wsmr.atec.list.datts-sv-e@mail.mil

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Marine Corps

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Marine Corps Systems Command (MARCORSYSCOM)
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2200 Lester Street
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Any information relating to Government contracts should be obtained through contracting officers.

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CONCLUDING MATERIAL

Custodians:
Army – SY
Navy – SH
Air Force – 11
DISA – DC5

Preparing activity:
DISA (DC5)
(Project EMCS-2022-022)

Review activities:
Army – AC, AM, AR, AT, AV, CE, CR, GL, MD, MI, PT, TE
Navy – AS, CG, EC, MC, OS, YD
Air Force – 13, 19, 22
DTRA – DS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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